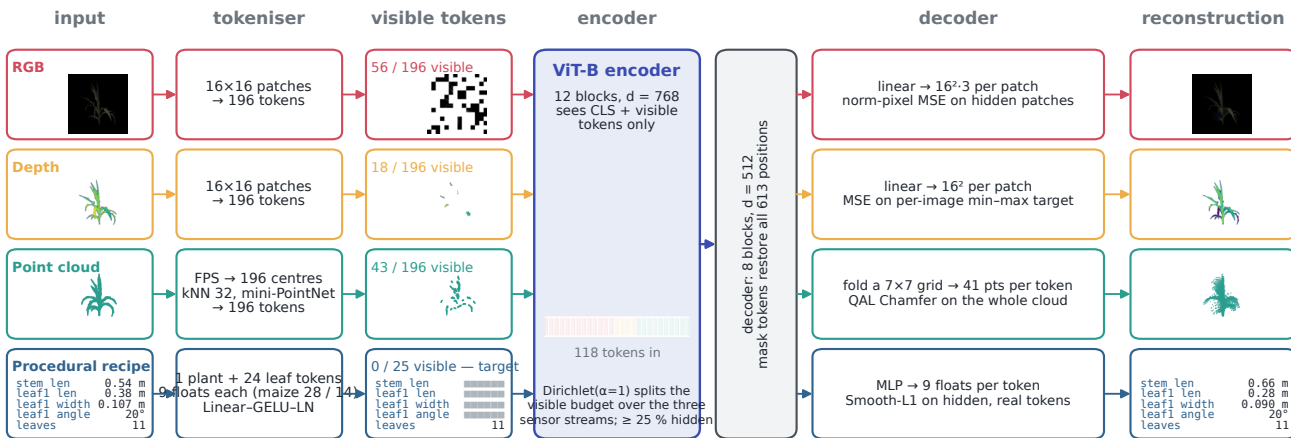
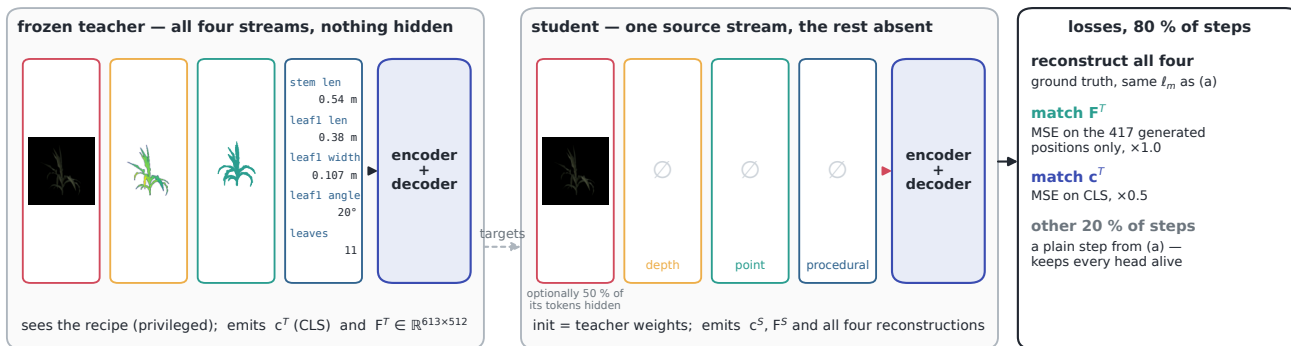


(a) Four-stream masked pretraining



The decoder layout never depends on the mask, so two forward passes with different visible sets give position-aligned features — the hook for (b).

(b) A vision-complete teacher and single-sensor students



all panels: held-out sorghum plant Sorghum_10_04 (data v2), real inputs, masks and model outputs; the maize model is identical apart from its recipe tokens (1 + 28 tokens of 14 floats)